Prospective low-prevalence evaluation

10,000 new URLs • 10.00% phishing • frozen models and thresholds

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
Weighted HYB-MS real-only	97.20%	86.66%	85.10%	85.87%	97.03%	91.34%	1.46%	14.90%
Weighted HYB-MS +10% GAN	97.84%	89.92%	88.30%	89.10%	97.45%	93.51%	1.10%	11.70%

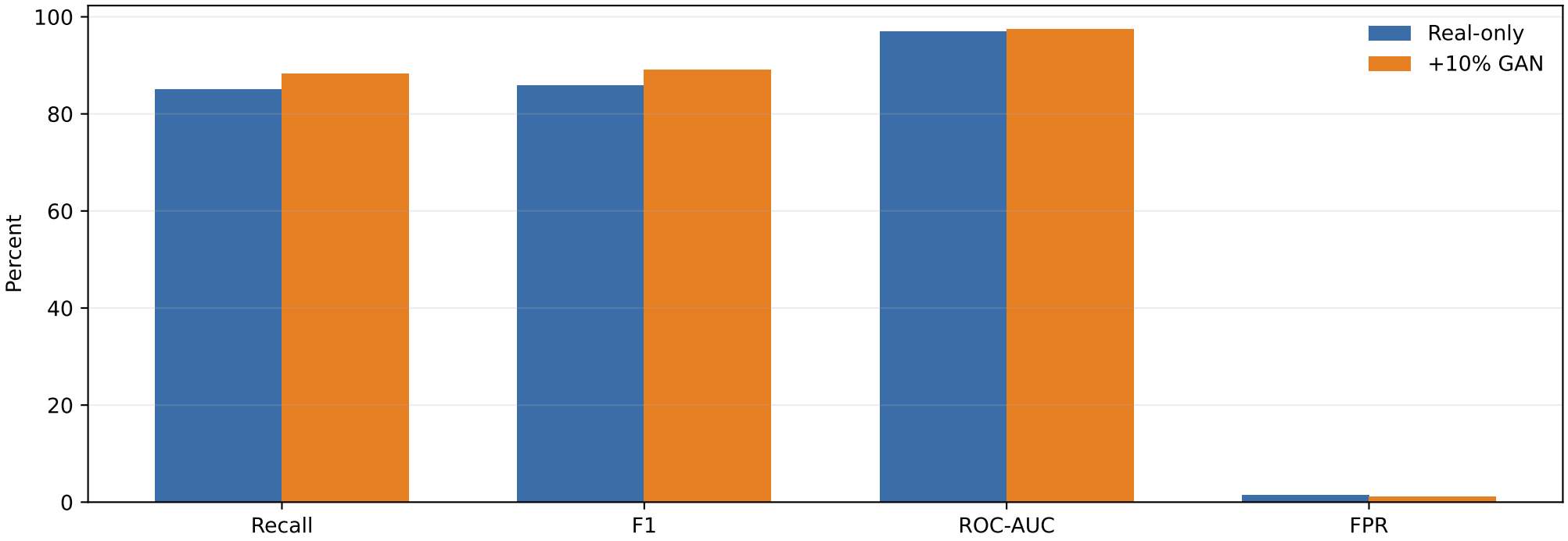
AUGMENTATION BENEFIT SUPPORTED

The +10% model improved recall, F1 and ROC-AUC without increasing FPR on this new cohort.

No training, calibration, checkpoint choice or threshold was changed after this dataset was frozen.

Dataset SHA-256: cec7f6b43b3ebcbd12e345833489f603071459a551e997da749390e69d1a4ffe

What the result means in browser-like traffic



Real-only false alerts per 10,000 visits: 131.0

+10% GAN false alerts per 10,000 visits: 99.0

This is an identity-disjoint, proxy-labelled evaluation. It is useful evidence, but it is still not a live production deployment test.